



NAMAL UNIVERSITY MIANWALI
Department of Computer Science

Database Systems Project Proposal
CCP - Milestone 1
CattleCare Pro: Livestock Breeding and Farm
Management Database System

Subject: Database Systems

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Submission Date: March 12, 2026

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Chapter 1

Introduction

Livestock farming plays an important role in the agricultural economy. Proper management of cattle records, breeding history, and financial transactions is essential for successful livestock farming businesses. However, many farms still rely on manual record keeping systems such as registers or spreadsheets, which often lead to data loss, inaccurate information, and difficulty in retrieving historical records.

The proposed system, **CattleCare Pro**, is a database-driven livestock breeding and farm management system designed to store and manage structured information related to cattle records, breeding history, purchase and sale transactions, birth and death records, and financial profit/loss records.

The primary objective of this project is to design a relational SQL database that allows efficient storage, retrieval, and records of livestock breeding data. The system will help farm owners manage cattle records effectively and generate useful insights for decision making.

Chapter 2

Problem Statement

Many livestock farms maintain cattle breeding and financial records manually. This traditional approach introduces several problems including difficulty in managing large volumes of data, loss of historical records, and inefficient data retrieval.

Major challenges faced by farm owners include:

- Difficulty maintaining complete cattle history records
- Lack of organized breeding information
- Difficulty tracking cattle birth and death records
- Poor management of cattle purchase and sale transactions
- Lack of accurate financial profit records
- Inefficient retrieval of cattle information

Without a properly structured database system, it becomes difficult to store and analyze livestock breeding data efficiently. Therefore, a relational SQL-based database system is required to manage cattle records, breeding history, and financial transactions in a structured and reliable way.

Chapter 3

Objectives

The main objectives of the proposed system are:

- Design a relational SQL database for livestock breeding management.
- Store detailed cattle records including identification number, age, and weight.
- Maintain breeding records information.
- Record cattle color information.
- Maintain death records if any cattle dies.
- Manage cattle purchase and sale transactions.
- Track financial information including purchase price, sale price, and profit.

Chapter 4

Scope

The scope of the proposed database system includes the following:

- Cattle registration and record management
- Management of breeding records
- Birth tracking
- Management of death records
- Cattle purchase and sale transaction management
- Financial profit/loss records
- Color and weight records
- Historical reporting
- User account and role management

The system focuses mainly on database schema design, relational data modeling, and SQL query operations rather than full application or hardware integration.

Chapter 5

Data Collection

Data required for the system will be collected from the following sources:

- Existing livestock farm records
- Purchase and sale transaction logs
- Breeding and birth history maintained by farm staff
- Manual data entry by the farmer or farm owner

The database will store structured information including cattle details, breeding records, birth and death data, purchase and sale transactions, financial records, and color-weight information of the cattle.

Chapter 6

Technology Stack

The following technologies will be used to implement the proposed database system:

- **Database Management System:** MySQL
- **Query Language:** SQL
- **Database Modeling Tool:** Draw.io or ERDPlus
- **Version Control:** GitHub
- **Development Tools:** MySQL Workbench

Chapter 7

System Functionality

The database system will provide the following functionalities:

1. Register and manage cattle information.
2. Store cattle breed, age, identification number,color, and weight.
3. Record cattle birth date information.
4. Maintain breeding records between cattle.
5. Track cattle purchase and sale transactions.
6. Store purchase price and sale price information.
7. Calculate profit or loss from cattle sales.
8. Maintain death records if cattle dies.

Chapter 8

User Classes

The system will support the following user roles:

8.1 Farm Owner/Administrator

- Full access to database records
- Manage cattle records
- Manage breeding and transaction records
- View cattle records
- Update weight and color information
- Record birth and death events

Chapter 9

Database Design Overview

The database system will consist of the following major entities:

- Cattle
- Breeding Records
- Birth Records
- Death Records
- Purchase Records
- Sale Records
- Weight Records
- Color Records
- Profit/Loss Records

These entities will be connected through relational tables using primary keys and foreign key relationships to ensure data integrity and consistency within the system.

Chapter 10

Task Division

The project tasks will be divided among team members as follows:

- **Muhammad Ali**

- Database schema design
- ER diagram creation
- GitHub repository management

- **Muhammad Ahmad**

- SQL query implementation
- Data insertion and testing

- **Aliya Ashraf**

- Documentation
- Database normalization
- SQL table implementation

Chapter 11

Conclusion

The proposed database system provides a structured solution for managing livestock breeding and farm management data. By implementing a relational SQL database, the system will enable efficient storage and retrieval of cattle records, breeding history, financial transactions, and color-weight information.

The database will help farm owner/administrator maintain accurate livestock records, track breeding lineage, monitor financial performance for better farm management decisions.

Chapter 12

References

- Elmasri, R., Navathe, S. (2016). *Fundamentals of Database Systems*.
- Database Systems Course Material, Namal University
- IEEE Software Requirements Specification Standard
- GitHub Repository: <https://github.com/aligoharcodepulse/CattleCare-Pro/>]

Appendix A: AI Prompts

Tool Used: ChatGPT

Purpose: Assistance in generating the project proposal structure and documentation.

Example Prompt:

”Generate a LaTeX database system proposal for a livestock breeding and farm management database including problem statement, objectives, scope, and database schema overview.”

Access Date: March 7th, 2026